L'électrochimiothérapie (ECT) dans le traitement des métastases osseuses : revue systématique de la littérature

Titre original : Electrochemotherapy in the Treatment of Bone Metastases: A Systematic Review

Auteurs : Bocchi MB, Meschini C et al. | **Revue :** *Journal of clinical medicine* (2023) | **PMID :** 37834793 | **DOI :** 10.3390/jcm12196150 | **Modalité :** Scanner TDM / Fluoroscopie

1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Correspond directement aux séances d'Électrochimiothérapie (ECT sous Bléomycine 30 mg IV avec générateur IGEA) réalisées au scanner Achard -1 à Cochin, seules ou combinées à la cryoablation.

2. Synthèse Clinique en Français

Résumé exécutif : Revue systématique évaluant l'efficacité et la sécurité de l'électrochimiothérapie (ECT à la bléomycine) dans les métastases osseuses appendiculaires, pelviennes et rachidiennes. L'ECT associe l'injection intraveineuse de bléomycine à des impulsions électriques réversibles qui perméabilisent la membrane des cellules tumorales sans détruire la trame osseuse ni les structures nerveuses adjacentes.

Analyse détaillée : Contrairement aux ablations thermiques (RF, cryoablation, micro-ondes), l'électrochimiothérapie (ECT) est une modalité non thermique basée sur l'électroporation réversible : des impulsions électriques de haut voltage ouvrent temporairement des nanopores membranaires, multipliant par plus de 1000 la pénétration intracellulaire de la bléomycine. Cette revue systématique démontre un contrôle antalgique majeur chez plus de 75–85 % des patients et une préservation unique de la résistance mécanique des travées osseuses, du cartilage articulaire et des gaines nerveuses, permettant de traiter des localisations complexes (pied, rachis, péri-articulaire).

3. Protocole Technique & Paramètres Opératoires

Positionnement parallèle des électrodes aiguilles sous guidage TDM selon la géométrie tumorale (espacement 1,5 à 2,5 cm) • Bolus IV de Bléomycine (15 000 UI/m² soit ~25–30 mg) • Fenêtre thérapeutique d'électroporation entre 8 min et 40 min post-injection (8 impulsions de 100 µs à 1000 V/cm par paire d'électrodes) sous AG avec curarisation.

4. Points Clés à Retenir pour la Pratique

- Respect absolu de la fenêtre pharmacocinétique : délivrance des impulsions électriques entre 8 et 40 minutes après le bolus IV de bléomycine.
- Absence d'effet thermique délétère : préserve la matrice extracellulaire osseuse, l'os sous-chondral, les vaisseaux et les nerfs périphériques.
- Indication de choix pour les métastases osseuses distales (pied, main), juxta-articulaires ou radiorésistantes (cancer du rein, mélanome).
- Synergie remarquable avec la cryoablation ('Cryo + ECT') ou la cimentoplastie pour les lésions volumineuses ostéolytiques.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Background and aim Cancers are one of the most frequent causes of death and disability in humans. Skeletal involvement has a major impact on the quality of life and prognosis of cancer patients. Electrochemotherapy is a palliative and minimally invasive oncologic treatment that was first used to treat subcutaneous nodules for malignant tumors. The aim of our review is to evaluate the results of electrochemotherapy in the treatment of

bone metastases. Methods A systematic review of the literature indexed in the PubMed MEDLINE and Cochrane Library databases using the search key words "electrochemotherapy" AND ("metastasis" OR "metastases") was performed. The Preferred Reporting Items for Systematic Reviews and MetaAnalyses was followed. Inclusion criteria were proven involvement of the appendicular skeleton in metastatic carcinoma or melanoma, through at least one percutaneous electrochemotherapy session on the metastatic bone lesion. The exclusion criterion was no skeletal metastatic involvement. Results Eight articles were finally included. We reached a population of 246 patients. The mean age and follow up were 60.1 years old and 11.4 months, respectively. The most represented primary tumor was breast cancer (18.9%). A total of 250 bone lesions were treated with electrochemotherapy. According to RECIST criteria, in our population we observed 55.5% stable diseases. The mean pre-electrochemotherapy VAS value was 6.9, which lowered to 2.7 after treatment. Adverse events occurred in 3.4% of patients. Conclusions Electrochemotherapy as a minimally invasive and tissue-sparing treatment should be considered for patients with no other alternative to obtain tumor control and improvement in quality of life.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

1. Introduction

Cancers are one of the most frequent causes of death and disability in humans. An estimated 19 million new cancer diagnoses are made each year. Among the most frequent forms are breast (11.7%), lung (11.4%), colorectal (10%), and prostate (7%) neoplasms [1]. Primary lesions of bone are very rare, representing less than 0.2% of all malignancies. Secondary lesions turn out to be more frequent, most often arising from lung, breast, prostate, kidney, and thyroid cancers; in fact, bone is the third most common site for metastatic disease [2]. Metastases can affect the entire skeleton, with higher incidence in the spine, pelvis, skull, ribs, and long bones such as the humerus and femur [3]. Skeletal involvement has a major impact on the quality of life and prognosis of cancer patients [4]. Patients with metastatic bone disease are evaluated by a team that includes oncologists, radiation oncologists, and orthopedists. The treatment of metastatic lesions depends on various factors including the patient’s life expectancy, general condition, number of metastases, progression of neoplastic disease, sensitivity to treatment, presence of a pathologic fracture, the patient’s quality of life [5,6], and the use of scores such as the Mirel Score [7]. There are various types of treatment for patients with bone metastases. If surgery is needed, several options are available, among which are resection and reconstruction with prosthesis [8], of which custom-made products are increasing in use, or prophylactic synthesis with an intramedullary nail [9]. Chemotherapy or single low-dose radiotherapy can be used as neoadjuvant or adjuvant therapies, with good results but also with higher complication rates [10]. Among pharmacological treatments, the use of inhibitors of osteoclast activity, such as bisphosphonates, and monoclonal therapy, such as denosumab, have had good results [11]. There are also minimally invasive treatments, which include embolization, thermal ablation therapy, high-intensity focused ultrasound, and electrochemotherapy (ECT) [12]. Electrochemotherapy is a palliative minimally invasive oncologic treatment that was first used to treat subcutaneous nodules for malignant tumors in the head and neck region. This technique uses a combination of electroporation and intravenous infusion of chemotherapeutic drugs, such as bleomycin and cisplatin. Electroporation increases the efficacy of drugs by opening cellular transmembrane channels, using permeability-enhancing pulses of electric current, enabling them to reach intracellular targets. The use of electric pulses also induces a transient reduction in tumor blood flow, causing the drug to be trapped in the tissue for a longer time. The cytotoxic action of electrochemotherapy also acts on the endothelial cells of tumor blood vessels, thus reducing the blood supply to the cancer and consequently causing a cascade of tumor cell death surrounding the vessels [13,14]. The main indications for the use of ECT include superficial tumors, superficial metastatic melanoma, breast cancer, head and neck skin tumors, non-melanoma skin cancers, and finally Kaposi sarcoma [15]. With the production of new types of instruments, the type of lesions that can be treated has increased to include deep lesions, although the latter are still being studied and explored [16]. Furthermore, in recent years, many studies on the use of ECT on bone metastases have been published, demonstrating the safety and efficacy of this technique, especially in the field of pain reduction [17] (Figure 1).

Unlike other techniques, such as radiotherapy and thermal ablation, ECT has been shown to not affect osteogenic activity or bone integrity [13].

Femur model showing electrode needle placement with a plastic mask used to hold the needles in the correct position during treatment.

The aim of our review is to evaluate the results of ECT in the treatment of bone metastases.

2. Materials and Methods

A systematic review of the literature indexed in the PubMed MEDLINE and Cochrane Library databases using the search key words “electrochemotherapy” AND (“metastasis” OR “metastases”) was performed in August 2023. To minimize the number of missed studies, only the “title/abstract” filter was applied to the search strategy. The bibliography of the selected studies was accurately searched by hand to identify further studies not found during the electronic search. No restrictions on the date of publication or language were applied. The title of the journal, name of authors, and supporting institutions were known at all stages.

The Preferred Reporting Items for Systematic Reviews and MetaAnalyses (PRISMA) was followed as reported in Figure 1. This study was not registered and therefore there is no registration number. This article adhered to the latest Preferred Reporting Items for Systematic Reviews and MetaAnalyses statement [18]. In order to be considered for this review, the articles needed to present some inclusion criteria: proven skeletal involvement in metastatic carcinoma or melanoma and at least one percutaneous ECT session on the metastatic bone lesion.

No skeletal metastatic involvement represented an exclusion criterion.

Abstracts and full texts were independently screened by two authors (R.V. and M.B.B.), and any discordance was solved by consensus with a third author (C.M.). The modified Coleman Methodology Score (mCMS) [19] was used to assess the methodological quality of the studies.

Each article was evaluated by two independent investigators (A.Z. and S.P.); in cases with more than a five-point difference between their ratings, the discrepancy was solved by consensus with a third author (A.P.). The mCMS ranges from 0 to 100 points, representing a well-designed study with no bias or confounding factors. All the selected studies were retrospectively analyzed by an author (M.B.B.) who then extracted and entered the data in an Excel worksheet. The collected data included main author, year of publication, article type, mCMS, patient age and gender, mean follow up, primary tumor, bone involvement, number of procedures, additional previous treatments, local tumor control, clinical outcomes, and potential complications. Local response to therapy was assessed using RECIST (Response Evaluation Criteria in Solid Tumors), which provides a simple and pragmatic methodology to evaluate the activity and efficacy of new cancer therapeutics in solid tumors, in this case for ECT. Pain levels were assessed through the Visual Analogue Scale (VAS), which measures pain intensity. Lastly, the data sheet was reviewed by two authors (R.V. and S.S.) who agreed on the extracted data.

3. Results

The searches resulted in 294 articles. Following the PRISMA flow chart [18], 16 articles were relevant to the general topic area and finally 8 were included in the review according to inclusion and exclusion criteria (Figure 2) (Table 1) [14,17,20,21,22,23,24,25]. Ranieri G et al. [23] did not exclusively report ECT treatment of skeletal metastases unlike the other authors. It was, however, possible to extrapolate data on their experience in the treating of bone metastases with ECT and thus include this work in our review.

mCMS: modified Coleman Methodology Score.

All studies considered were prospective clinical studies except for the works of Cornelis FH et al. [21], Gasbarrini A et al. [22], and Deschamps F et al. [25], which are a case report, a preliminary study, and a retrospective study, respectively. According to the mCMS evaluation, the mean score of the studies reached was 70 points (44–94 points), which represents a suboptimal study design.

We reached a population of 246 patients, of which 105 were males and 141 were females. The mean age of the study population was 60.1 years old. The mean follow up was 11.4 months (Table 1).

The most-represented primary tumor was breast cancer (18.9%), followed by kidney (16%), lung (10.7%), colon (9.1%), thyroid (6.2%), bladder (4.1%), soft tissue sarcoma (3.3%), endometrium (2.9%), melanoma and prostate (both 2%), and thymoma (1.2%). Finally, in 18.5% and 4.9% of cases the primary tumor was either different from the above or unknown, respectively (Table 2).

STS, soft-tissue sarcoma; Unk, unknown.

A total of 250 bone lesions were treated with ECT. Most of the metastases were in the limbs (54.8%) (upper 20% and lower 80%), while the remaining were located in the trunk (45.2%). Further in detail, the skeletal segments most affected by metastases were the femur (30.1%) and the pelvic ring (26.7%). However, 20.8% of metastatic lesions had costo-vertebral localization instead. More specifically, we are aware that 85% of these were vertebral metastases, since it was specified by some of the authors [22,23,24,26] (Table 3).

Many patients underwent previous treatment for bone metastases before ECT: 78 patients received chemotherapy (31.7%), 96 received radiotherapy (39%), 23 received hormone therapy (9.3%), 3 received lesion embolization (1.2%), and finally 55 patients received further unspecified treatments (22.3%).

4. Discussion

To the best of the authors' knowledge, this is the first systematic review specifically on the use of ECT for the treatment of bone metastases. However, a systematic review of the literature on electrochemotherapy in solid abdominal organ and bone tumors was recently published [26]. The treatment of bone metastases with ECT is discussed but is not the mainstay of the review; moreover, the works cited are a small part of what has been published on the topic in the literature.

Metastatic bone disease is a significant healthcare issue, in fact almost 75% of metastatic bone lesions are symptomatic and require local treatment to improve patients' outcome and prevent local disease progression [27]. To date, there is no gold-standard treatment for bone metastases. Radiotherapy is the most used local therapy for bone metastases, with estimated rates of pain relief reported in the literature between 50% and 80%. To prove this, all patients treated by Deschamps et al. [25] for instance underwent prior radiotherapy for metastatic epidural spinal cord compression, with, unfortunately, no resolution. This treatment option, however, is not without side effects; therefore, further recent technologies have been used [28,29,30]. The common endpoint to all ablation techniques is the induction of the largest possible thermal necrosis of the target lesion to destroy periosteal nociceptors and reduce tumor size. Compared to other ablation treatments, ECT is minimally invasive, tissue-sparing, and repeatable [31]. Furthermore, ECT ensures good pain control, avoiding negative impacts on quality of life [32].

Primary tumor distribution almost completely overlaps across the literature [1] (breast, kidney, lung, and colon); conversely, the localization of bone metastases does not. According to the data collected, the appendicular skeleton seems to be slightly more involved (54.8%) compared to spine (20.8%) and pelvic (26.7%) localizations. However, this distribution seems to agree with the indications for ECT as an adjuvant therapy in pathological/impending long-bone fractures requiring nailing (femur and tibia) and for lesions that are difficult to access with surgery (pelvis and spine).

In line with data collected in the literature, the radiological response to ECT treatment according to RECIST criteria was mostly a "stable disease" (55.5%); conversely, the number of patients with "disease progression" was negligible (6.4%). In addition, regardless of the classification system used to evaluate the radiological response to ECT (RECIST or MDA [14]), "stable disease" appears to be the most frequent response. In conclusion, local control in bone metastases is achieved with percutaneous ECT.

Pain proved to be a valid clinical outcome indicator, especially in patients who underwent ECT palliative treatments. All the articles included in our review showed a significant reduction in pain symptoms after ECT (VAS values lowered from 6.9 pre ECT to 2.7 post ECT). The decrease appears to be around 60%, showing that ECT obtains results in terms of pain reduction comparable to radiotherapy. Campanacci L et al. [17,20]

also observed a reduction in pain even in the absence of an objective radiological response.

ECT does not induce bone necrosis on healthy tissue and does not damage the mineral structure of the bone and its regenerative capacity. Therefore, in case a fracture of the ECT-treated bone occurs, bone healing is possible with usual fracture callus quality and time [33]. In this regard, Campanacci L et al. [17] and Cevolani L et al. [24] studied a slice of the population affected by osteolytic metastasis with either a pathological fracture or an impending fracture, in which there was an indication for preventing nailing fixation. The authors performed tumor electroporation followed by intramedullary nailing: their clinical and radiological results suggested that combining the two procedures does not impact the overall response to the orthopedic treatment. However, Cevolani L et al. [24] suggested that it might be useful to carry out additional local treatments such as radiotherapy or embolization afterward.

Furthermore, since electropermeabilization is a universal phenomenon, any kind of tumor can be treated by ECT, regardless of its usual sensitivity to bleomycin or cisplatin [34].

Regarding the ECT treatment of bone metastases with spinal localization, we found conflicting opinions. Bianchi G et al. [14] considered spine metastases to limit ECT treatment due to the proximity of the spinal cord and technical limitations of electrode positioning. However, it must be said that Bianchi's group's experience is certainly the oldest one, and possibly their results and therefore their conclusions may also be influenced by this. In contrast, Gasbarrini A et al. [22], Cornelis FH et al. [21], and not least Deschamps et al. [25] treated spinal metastases only, proving ECT to be a novel and adjuvant treatment of spinal metastases, regardless of the histological types.

5. Conclusions

Despite the limitations, we can conclude that electrochemotherapy, as a minimally invasive and tissue-sparing treatment, should be considered as an alternative, either alone or with further treatments, for patients with bone metastases to obtain tumor control and improvements in their quality of life.

Author Contributions

(I) Conception and design: M.B.B. and R.V.; (II) administrative support: G.M.; (III) provision of study materials or patients: A.P. and A.Z.; (IV) collection and assembly of data: M.S.O., C.M., M.R.M. and S.P.; (V) data analysis and interpretation: M.B.B. and R.V.; (VI) manuscript writing: all authors; (VII) final approval of manuscript: all authors. All authors have read and agreed to the published version of the manuscript.

Conflicts of Interest

The authors declare no conflict of interest.

PRISMA 2020 Checklist Statement

The authors have read the PRISMA 2020 Checklist, and the manuscript was prepared and revised according to the PRISMA 2020 Checklist.